

1	YTSS/2025/I/06
	$7^b = 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a}$. Express b in terms of a . [2]
2	YTSS/2025/I/07
	Simplify $\left(\frac{x^9 y^3 z^0}{27}\right)^{\frac{1}{3}} \times 3x^{-2}$. [2]
3	SCGS/2025/I/02
	(a) Simplify $\sqrt[3]{\frac{x^9}{27y^6}}$. [1]
	(b) It is given that $25 \times 7^6 + 24 \times 49^3 = 7^{2n}$. Use the laws of indices to find the value of n . [2]
4	SCGS/2025/I/05
	$P = 2^3 \times 3^2 \times 11$. $Q = 2^k \times 11^2$, where k is a positive integer less than 3.
	(a) Write down, in terms of k , the highest common factor of P and Q . Give your answer as a product of its prime factors. [1]
	(b) $P \div \frac{m}{n}$ is a perfect cube, where m and n are prime numbers. Find the value of m and of n . [2]
5	SST/2025/I/03(a)
	(a) Given that $2^{m-1} \times 3^m = 1$, find the value of 36^m . [2]
6	GMSS/2025/I/05
	Three strings of lengths 24 cm, 36 cm and 90 cm are cut into shorter equal lengths with no remainder.
	(a) Find the largest possible length that each string can be cut into. [1]
	(b) Hence, find the total number of strings of such length. [1]
7	GMSS/2025/I/22
	Write the following numbers in order of size, starting with the smallest. 32^{100} , 25^{250} and 81^{125} Show your working clearly. [2]

8	ACSBR/2025/I/02
	(a) Simplify $\left(\frac{a^3b^9}{125}\right)^{-\frac{1}{3}}$. [2] (b) $9 \div 27^2 \times \frac{1}{81^{-k}} = \sqrt{3}$ Use the laws of indices to find the value of k. Show your workings. [3]
9	PHS/2025/I/09
	The distance between Earth and Venus is 150.4 million kilometres. (a) Express 150.4 million in standard form. [1] (b) A space shuttle can travel 7500 metres in 1 second. (i) Find the speed, in kilometres per hour, of the space shuttle. [1] (ii) Find the number of days the space shuttle will take to travel from Earth to Venus. Give your answer to the nearest whole number. [2]
10	GESS/2025/I/01
	(a) Calculate $\frac{33.17 \times 55.2^{\frac{1}{3}}}{-(-1) + \sqrt[3]{647.9}}$, giving your answer correct to 1 significant figure. [1] (b) Convert 3 m/s to km/h. [1]

Solution

1	$7^b = 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a} + 7^{2a}$ $7^b = 7(7^{2a}) \text{---[M1]}$ $7^b = 7^{2a+1}$ $b = 2a+1 \text{---[A1]}$
2	$\left(\frac{x^9 y^3 z^0}{27} \right)^{\frac{1}{3}} \times 3x^{-2}$ $= \frac{x^3 y}{3} \times \frac{3}{x^2} \text{---[M1]}$ $= xy \text{---[A1]}$
3	(a) $\frac{x^3}{3y^2}$ (b) $25 \times 7^6 + 24 \times 49^3 = 7^{2n} \quad \text{Alt} \quad 25 \times 7^6 + 24 \times 49^3 = 7^{2n}$ $25 \times 7^6 + 24 \times (7^2)^3 = 7^{2n} \quad 25 \times (7^2)^3 + 24 \times 49^3 = 7^{2n}$ $25 \times 7^6 + 24 \times 7^6 = 7^{2n} \quad 25 \times 49^3 + 24 \times 49^3 = 49^n$ $7^6 (25 + 24) = 7^{2n} \quad 49^3 (25 + 24) = 49^n$ $7^6 (25 + 24) = 7^{2n} \quad 49^3 (49) = 49^n$ $7^6 (49) = 7^{2n} \quad n = 3 + 1$ $7^{6+2} = 7^{2n} \quad n = 4$ $8 = 2n$ $n = 4$
4	(a) $2^k \times 11$ (b) $m = 11$ $n = 3$
5	$2^{m-1} \times 3^m = 1$ $2^m \times 3^m = 2$ $6^m = 2$ $36^m = 2^2$ $36^m = 4$
6	(a) Largest possible length = 6 cm (b) $\frac{24}{6} + \frac{36}{6} + \frac{90}{6} = 4 + 6 + 15 = 25$

7	$32^{100}, 25^{250} \text{ and } 81^{125}$ $= 2^{5 \cdot 100}, 5^{2 \cdot 250}, 3^{4 \cdot 125}$ $= 2^{500}, 5^{500}, 3^{500}$ Ascending order: $32^{100}, 81^{125}$ and 25^{250}.														
8	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; text-align: center;">(a)</td><td style="width: 90%;">$\left(\frac{a^3 b^9}{125}\right)^{\frac{1}{3}}$</td></tr> <tr> <td></td><td>$= \frac{5}{ab^3}$</td></tr> <tr> <td style="text-align: center;">(b)</td><td>$9 \div 27^2 \times \frac{1}{81^{-k}} = \sqrt{3}$</td></tr> <tr> <td></td><td>$3^2 \div 3^6 \times \frac{1}{3^{-4k}} = 3^{\frac{1}{2}}$</td></tr> <tr> <td></td><td>$3^2 \div 3^6 \times 3^{4k} = 3^{\frac{1}{2}}$</td></tr> <tr> <td></td><td>$2 - 6 + 4k = \frac{1}{2}$</td></tr> <tr> <td></td><td>$k = 1.125 \text{ or } k = \frac{9}{8}$</td></tr> </table>	(a)	$\left(\frac{a^3 b^9}{125}\right)^{\frac{1}{3}}$		$= \frac{5}{ab^3}$	(b)	$9 \div 27^2 \times \frac{1}{81^{-k}} = \sqrt{3}$		$3^2 \div 3^6 \times \frac{1}{3^{-4k}} = 3^{\frac{1}{2}}$		$3^2 \div 3^6 \times 3^{4k} = 3^{\frac{1}{2}}$		$2 - 6 + 4k = \frac{1}{2}$		$k = 1.125 \text{ or } k = \frac{9}{8}$
(a)	$\left(\frac{a^3 b^9}{125}\right)^{\frac{1}{3}}$														
	$= \frac{5}{ab^3}$														
(b)	$9 \div 27^2 \times \frac{1}{81^{-k}} = \sqrt{3}$														
	$3^2 \div 3^6 \times \frac{1}{3^{-4k}} = 3^{\frac{1}{2}}$														
	$3^2 \div 3^6 \times 3^{4k} = 3^{\frac{1}{2}}$														
	$2 - 6 + 4k = \frac{1}{2}$														
	$k = 1.125 \text{ or } k = \frac{9}{8}$														
9	(a) $150.4 \times 10^6 = 1.504 \times 10^8 \text{ km}$ (bi) $\frac{7500 \div 1000}{1 \div 3600}$ $= 27000 \text{ km/h}$ (bii) $\frac{1.504 \times 10^8}{27000} \div 24$ 232 days														
10	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; text-align: center;">1(a)</td><td style="width: 90%;"> $\frac{33.17 \times 55.2^{\frac{1}{3}}}{-(-1) + \sqrt[3]{647.9}}$ $= 13.08359745$ $= 10$ </td></tr> <tr> <td style="text-align: center;">1(b)</td><td> $\frac{3m}{s} = \frac{0.003km}{\frac{1}{3600}h}$ $= 10.8km / h$ </td></tr> </table>	1(a)	$\frac{33.17 \times 55.2^{\frac{1}{3}}}{-(-1) + \sqrt[3]{647.9}}$ $= 13.08359745$ $= 10$	1(b)	$\frac{3m}{s} = \frac{0.003km}{\frac{1}{3600}h}$ $= 10.8km / h$										
1(a)	$\frac{33.17 \times 55.2^{\frac{1}{3}}}{-(-1) + \sqrt[3]{647.9}}$ $= 13.08359745$ $= 10$														
1(b)	$\frac{3m}{s} = \frac{0.003km}{\frac{1}{3600}h}$ $= 10.8km / h$														